

Main idea

- Keep the RealSense camera connected to the main PC.
- Do the background removal, depth work, and cone warp on the main PC.
- Send only the finished warped video over Wi-Fi.
- Use a second laptop at the display for the first test.
- Do not buy a Raspberry Pi until the laptop test works.

Current program

- `live\_view.py` opens the camera with OpenCV.
- The regular preview shows the camera image.
- The fullscreen display runs `\_apply\_warp()` first.
- `\_apply\_warp()` removes the background, centers the person, applies the cone map, and adjusts the color.
- That finished frame is what we should send wirelessly.
- RealSense depth processing is still future work.

Camera -> main PC processing -> cone warp -> Wi-Fi -> second laptop -> HDMI display

Research notes

- Pepper's Cone and Virtual Showcase both show that the picture must be warped for the reflector [1][2][3].
- Disney's patents show that the reflector angle, material, tension, and room lighting also matter [4][5][6].
- RoomAlive used depth cameras for tracking and calibration [7].
- Holoportation used several RGB-D cameras, but that system is much larger than what we need [8].
- Our first depth goal should be one clean person cutout from one RealSense camera.
- The RealSense D435 should stay plugged into the main PC through USB 3 [9].
- WebRTC and NDI are possible ways to send the finished video with low delay [10][11][12].

Test notes

Use two laptops:

- Laptop 1: camera and all image processing.
- Laptop 2: receives the video and shows it fullscreen on the 72-inch display.

Compare:

- Direct HDMI
- MJPEG
- H.264 with WebRTC
- NDI, if available
- Wireless HDMI, if available

Record:

- Delay from movement to the display
- Frames per second
- Dropped or repeated frames
- Network use
- CPU and GPU use
- Image quality around the person

A 240 fps phone recording can help measure the delay. Run each test three times for ten minutes. Keep the camera position, lighting, video size, and test video the same.

Current decision

- Try MJPEG first because it is the easiest test.
- Try H.264 with WebRTC next.
- Compare both against direct HDMI.
- Aim for a stable 30 fps.
- Start with a goal of less than 150 ms delay. Less than 100 ms would be better.
- Wait to buy a Raspberry Pi until we know what the receiver needs.

Links

1. [Pepper's Cone paper](<https://grail.cs.washington.edu/wp-content/uploads/2017/10/luo2017pca.pdf>)
2. [The Virtual Showcase](<https://doi.org/10.1109/38.963460>)
3. [Virtual Showcase image warping]([https://doi.org/10.1016/S0097-8493\(03\)00087-6](https://doi.org/10.1016/S0097-8493(03)00087-6))
4. [Disney anamorphic Pepper's Ghost patent](<https://patents.google.com/patent/US8262226B2/en>)
5. [Disney multiview Pepper's Ghost patent](<https://patents.google.com/patent/US8692738B2/en>)
6. [Disney reflector tensioning patent](<https://patents.google.com/patent/US9625732B1/en>)
7. [RoomAlive](<https://www.microsoft.com/en-us/research/publication/roomalive-magical-experiences-enabled-by-scalable-adaptive-projector-camera-units/>)
8. [Holoportation](<https://www.microsoft.com/en-us/research/publication/holoportation-virtual-3d-teleportation-in-real-time/>)
9. [Intel RealSense D435 specifications](<https://www.intel.com/content/www/us/en/products/sku/128255/intel-realsense-depth-camera-d435/specifications.html>)
10. [Salsify low-latency video paper](<https://www.usenix.org/conference/nsdi18/presentation/fouladi>)
11. [WebRTC transport standard](<https://www.rfc-editor.org/info/rfc8835>)
12. [NDI technical requirements](<https://docs.ndi.video/all/developing-with-ndi/ndi-certified/certification-guidelines/technical-requirements>)